

Instruction:

For the homework, a dataset will be provided to you on Canvas, and you need to download it for use. A final Jupyter Notebook file (.ipynb) needs to be submitted as your answers to the homework questions.

The Jupyter Notebook you submit should have:

- All code (data loading, preprocessing, models, evaluation)
- Explanatory comments and short written answers to conceptual questions
- Clear tables/prints of evaluation metrics and at least one plot (e.g. ROC curve or feature importance bar chart).

You are strongly encouraged to work on this problem set together in the form of a study group. However, everyone should submit his/her individual write-up.

Homework: Predicting Firm Default Risk with Supervised Learning**Background**

You are working as a data scientist in a credit risk team of a commercial bank. The bank lends to small and medium-sized firms and wants to predict which firms are likely to default on their loans in the next year. Accurate predictions help the bank price risk, set credit limits, and comply with capital regulations.

You have access to a firm-level panel collapsed to one row per firm with the following variables:

- firm_id – firm identifier
- sector – industry sector (e.g. manufacturing, services, retail)
- region – region code (e.g. East, Central, West)
- age – firm age in years
- employees – number of employees
- revenue – annual revenue (in million local currency units)
- assets – total assets (in million)
- leverage – total debt / total assets
- current_ratio – current assets / current liabilities
- interest_coverage – EBIT / interest expenses
- export_share – share of revenue from exports (0–1)
- previous_default – indicator =1 if the firm ever defaulted in the past, 0 otherwise
- default_next_year – target variable, =1 if the firm defaults in the next 12 months, 0 otherwise

Your task is to build and evaluate supervised learning models to predict default_next_year.

A dataset file “firms_default.csv” is provided.

Tasks

1. Problem setup (conceptual)

- 1.1. Is this a classification or regression problem? Explain briefly.
- 1.2. Discuss two economic reasons why a bank might care more about false negatives (predicting no default when the firm actually defaults) than about false positives.
- 1.3. Suppose the unconditional default rate in the sample is 5%. Why is it not enough to just build a model that always predicts “no default”?

2. Data preparation

Using Python (pandas, scikit-learn):

- 2.1. Load the data and report:
 - a) Number of observations
 - b) Number of variables
 - c) Proportion of firms that default next year
- 2.2. Create dummy variables for sector and region. Explain why we need to convert these categorical variables before feeding them into most ML models.
- 2.3. Split the data into train (70%) and test (30%) sets using `train_test_split`. Use `random_state=123` for reproducibility. Do not shuffle by `firm_id`, just use the standard random split.

3. Baseline model: Logistic regression

- 3.1. Estimate a logistic regression model with `default_next_year` as the dependent variable and the other variables (including dummies) as predictors.
- 3.2. On the test set, report:
 - a) Accuracy
 - b) Precision, recall, F1-score (for the “default = 1” class)
 - c) ROC-AUC
- 3.3. Interpret two coefficients in economic terms (e.g., leverage, `previous_default`): what is their sign, and does it match your priors?

4. Machine learning models: Tree-based methods

4.1. Fit a Random Forest classifier using the same predictors. Use a reasonable default (e.g. `n_estimators=200`, `max_depth` left as default).

4.2. Evaluate the Random Forest on the test set using the same metrics as in 3.2. Compare performance with logistic regression. Which model performs better on ROC-AUC and recall for defaults?

4.3. Plot and briefly discuss the feature importances from the Random Forest. Which *three* variables appear most important? Are they economically plausible?

4.4. Perform a simple hyperparameter search (e.g., grid search over `max_depth` and `min_samples_leaf`) using cross-validation on the training set. Does performance on the test set improve?

5. Economic interpretation and policy implications

5.1. Suppose the bank uses your model to approve loans only for firms whose predicted default probability is below 10%. Explain qualitatively how this might affect:

- a) Access to credit for young vs. old firms
- b) Sectoral credit allocation (e.g. manufacturing vs. services)

5.2. Discuss one potential concern about using such a model from the perspective of:

- a) Risk management (e.g. stability when the macro environment changes)
- b) Fairness / regulatory compliance (e.g. disparate impact across regions or sectors)

5.3. How could you evaluate whether the model is robust over time if you receive new data every year?